

Experimental overview

A total of 15 crabs consisting of 12 hairy shore crabs and 3 European green crabs were used for this study. Crabs were assigned to one of three salinity treatments consisting of 15 ppt, 35 ppt, or 45 ppt seawater for a period of three weeks. Each salinity treatment started with one European green crab and four hairy shore crabs. Crab weight, righting time, and metabolic activity was measured weekly throughout the duration of the experiment. Hemolymph extractions were also performed during weeks one and three to determine osmolality.

Righting time and weight

After a crab was weighed it was placed dorsal side down on a flat surface and the time required to right itself was recorded.

Hemolymph extractions

Hemolymph was extracted from crabs by inserting a small needle into the joint of a crab leg. The extracted hemolymph was then analyzed using an osmometer.

Respiration rates

Metabolic activity was measured using a resazurin assay. Individual crabs were placed into separate chambers with enough resazurin to submerge them completely, along with a blank chamber containing no crab. Aliquots of 200 ul were collected at 0, 30, and 60 minute time intervals and transferred to a 96 well plate. Fluorescence was measured using a spectrophotometer plate to quantify resorufin production. Afterwards, the crabs were rinsed and placed back into their respective salinity conditions.